0.5 Project Scope Sheet

Oct 24, 2026 · @Batista

Baseline A · Azores Express

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group ___
Member 1 (name · student number)	_____ · _____
Member 2	_____ · _____
Member 3	_____ · _____
Member 4	_____ · _____
Baseline	A · Azores Express: container ship, 1,300 TEU, dual-fuel HFO/LNG, Lisbon – Ponta Delgada – Praia da Vitória – Lisbon, weekly
Specialization 1	SP__ · _____
Specialization 2	SP__ · _____
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group___ _Project.zip (for example Group3_Project.zip)
Presentation	Dec 18, 2026, 15 minutes per group

How the work is split. The in-class exercises are individual practice of each method on the reference ship (the 37,600 t oil/chemical tanker) and are not graded. This project is your group's work, done outside class: you apply each method to the Azores Express and complete the full scope of every deliverable below. A report that only repeats the class exercises does not meet the scope.

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D1	Design Basis	Ex. 1.2 does the owner's brief add up, classify the ship · Ex. 1.3 design phases and documents	1,300 TEU at 10 t per TEU on average (13,000 t cargo) · 18–19 kn · weekly round voyage · limits at Lisbon, Ponta Delgada, Praia da Vitória (and Leixões), each with its source · environmental targets · every requirement tagged Given / Assumed / To be checked	
D2	Reference Ship Database	Ex. 1.4 first Δ, L, B, T, D from the App. A regressions, with 95% bands	At least 10 gearless container ships of about 900–1,800 TEU, recent · sources for every ship · your own regressions and bounds · Appendix A.3	
D3	Ship Model	Ex. 1.5 displacement equation · Ex. 2.3 main dimensions · Ex. 2.4 power · Ex. 3.1 weights · Ex. 3.2 capacity · Ex. 3.3 engine and propeller	Volume-carrier sequence (Table 1.9) · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · dual-fuel machinery · LNG storage aft in the weights · TEU capacity check · closed weight balance	

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D4	Voyage and Financial Model	Ex. 3.4 round voyage and fuel · Ex. 4.1 building cost · Ex. 4.2 RFR and NPV	One round voyage a week: sea and port time, LNG and HFO use, fuel and port costs, freight revenue · building cost, financing, RFR and NPV · ship model outputs as input	
D5	Validation Note	No class exercise: done by the group	Your model against the reference ships: L, B, T, Δ, lightship, power · error per quantity · model corrected where needed	
D6	Optimization Report	Ex. 5.1 synthesis model · Ex. 5.2 optimization (demonstration + individual step) · Ex. 5.3 multi-objective trade-off	Minimum propulsion power that keeps the weekly schedule · justified objective · active constraints · optimum against the requirements	
D7	Compliance Sheet	Ex. 4.3 GM, freeboard and attained EEDI	EEDI and CII as design targets · ECA near the mainland (Tier III, 0.10% S) · freeboard · stability with the deck stacks · draught at every port	
D8	General Arrangement and Hull Form	Ex. 5.4 sectional area curve and LCB · Ex. 6.1 sections, bow and stern · Ex. 6.2 GA and capacity plan	Bays, rows and tiers · forward superstructure · LNG storage aft · midship section · A3 sketch (profile and midship section) · hydrostatics check	

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D9	Specialization Documents (2)	No class exercise: self-study decks	One calculation note per specialization, with its required drawings (see Specialization 1 and 2 above)	
D10	Datasheet and Presentation	No class exercise	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes	

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.

Baseline B · Azores Feeder

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group ___
Member 1 (name · student number)	_____ · _____
Member 2	_____ · _____
Member 3	_____ · _____
Member 4	_____ · _____
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP__ · _____

Field	Value
Specialization 2	SP__ · _____
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group___ _Project.zip (for example Group4_Project.zip)
Presentation	Dec 18, 2026, 15 minutes per group

How the work is split. The in-class exercises are individual practice of each method on the reference ship (the 37,600 t oil/chemical tanker) and are not graded. This project is your group's work, done outside class: you apply each method to the Azores Feeder and complete the full scope of every deliverable below. A report that only repeats the class exercises does not meet the scope.

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D1	Design Basis	Ex. 1.2 does the owner's brief add up, classify the ship · Ex. 1.3 design phases and documents	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked	
D2	Reference Ship Database	Ex. 1.4 first Δ, L, B, T, D from the App. A regressions, with 95% bands	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds	

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D3	Ship Model	Ex. 1.5 displacement equation · Ex. 2.3 main dimensions · Ex. 2.4 power · Ex. 3.1 weights · Ex. 3.2 capacity · Ex. 3.3 engine and propeller	Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance	
D4	Voyage and Financial Model	Ex. 3.4 round voyage and fuel · Ex. 4.1 building cost · Ex. 4.2 RFR and NPV	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input	
D5	Validation Note	No class exercise: done by the group	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed	
D6	Optimization Report	Ex. 5.1 synthesis model · Ex. 5.2 optimization (demonstration + individual step) · Ex. 5.3 multi-objective trade-off	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements	

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)	Status
D7	Compliance Sheet	Ex. 4.3 GM, freeboard and attained EEDI	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port	
D8	General Arrangement and Hull Form	Ex. 5.4 sectional area curve and LCB · Ex. 6.1 sections, bow and stern · Ex. 6.2 GA and capacity plan	Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · hydrostatics check	
D9	Specialization Documents (2)	No class exercise: self-study decks	One calculation note per specialization, with its required drawings (see Specialization 1 and 2 above)	
D10	Datasheet and Presentation	No class exercise	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes	

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.

Specializations to copy into the sheets

As on 0.2 Ship Design Project, slide 7. Each has its own self-study deck.

Code	Specialization	Must demonstrate
SP01	Longitudinal Strength Calculation	Loads and bending moments · section modulus check · midship section drawing
SP02	Hull–Propeller–Engine Matching	Optimum B-series propeller · Engine matching · PTO/PTI modes · Stern drawing
SP03	Electrical Load Balance	Consumers per operating mode · generator sizing · one-line diagram
SP04	Detailed Cost Estimation	Cost breakdown by group · unit cost sources · link to the general arrangement
SP05	Parametric Hull Shape Generation	Parametric hull definition · coupling to the optimizer · lines plan
SP06	Engine Room and Steering Room Arrangement	Machinery layout · rudder and steering gear · arrangement drawing
SP07	Preliminary Stability Calculation	Loading conditions · GZ curves against criteria · tank and capacity plan
SP08	Hull Shape Optimization	Hull variables and objective · resistance gain · initial and optimized lines
SP09	Damage Probabilistic Calculations	Baseline A only · watertight subdivision · A ⩾ R · subdivision plan
SP10	Detailed Emissions Calculation	Fuel use per mode · CO2, SOx, NOx per year · machinery arrangement
SP11	Advanced Dimensional Analysis	Regression equations for dimensions and ratios · fit quality and validity range
SP12	Alternative Fuel and Energy Storage	Energy per voyage · tank or battery sizing · rules check